

APPROVAL AND SCOPE

Design validation record

Executed laboratory results for prototype pack AQL-P3-017. This signed report is primary test evidence. Inspection after disassembly is recorded separately in AQL-QIR-221.

Table 1. Approval

Role	Name / status
Test engineer	M. Chen - signed 2026-07-19
Independent reviewer	L. Ortega - signed 2026-07-20
Test article	AQL-P3-017
Laboratory	North Cell DV Laboratory

Scope boundary. A PASS in this report applies only to the recorded test article and method. Later destructive-inspection observations remain authoritative for conditions they directly inspect.

TR-EL-01 VOLTAGE AND ACCURACY

TR-EL-01 - Electrical measurement characterization

Table 2. Pack monitoring sweep

Set point	Displayed voltage	Coverage result	Instrument
380 V	380.3 V	Below 400 V lower bound	DMM-22
400 V	400.2 V	Required lower bound	DMM-22
800 V	799.8 V	Required upper bound	DMM-22
820 V	819.6 V	Above 800 V upper bound	DMM-22

Table 3. Cell-voltage error by chamber point

Temperature	Worst absolute error	Acceptance	Coverage note
+25 deg C	4.2 mV	<= 5 mV	Tested
+60 deg C	4.8 mV	<= 5 mV	Tested upper endpoint
-20 deg C	Not measured	No result	Cold chamber unavailable

Result. Pack monitoring covers 380-820 V. Cell accuracy is demonstrated at +25 and +60 deg C; the required -20 deg C endpoint was not tested.

TR-SF-02 CRASH RESPONSE

TR-SF-02 - Frontal-impact electrical response

Table 4. Event timing and discharge

Event	Measured value	Requirement	Result
Crash input asserted	0.0 ms	Reference	Recorded
Both main contactors open	14.6 ms	<= 20 ms	PASS
DC bus at 4.20 s	48 V	< 60 V by 5.0 s	PASS

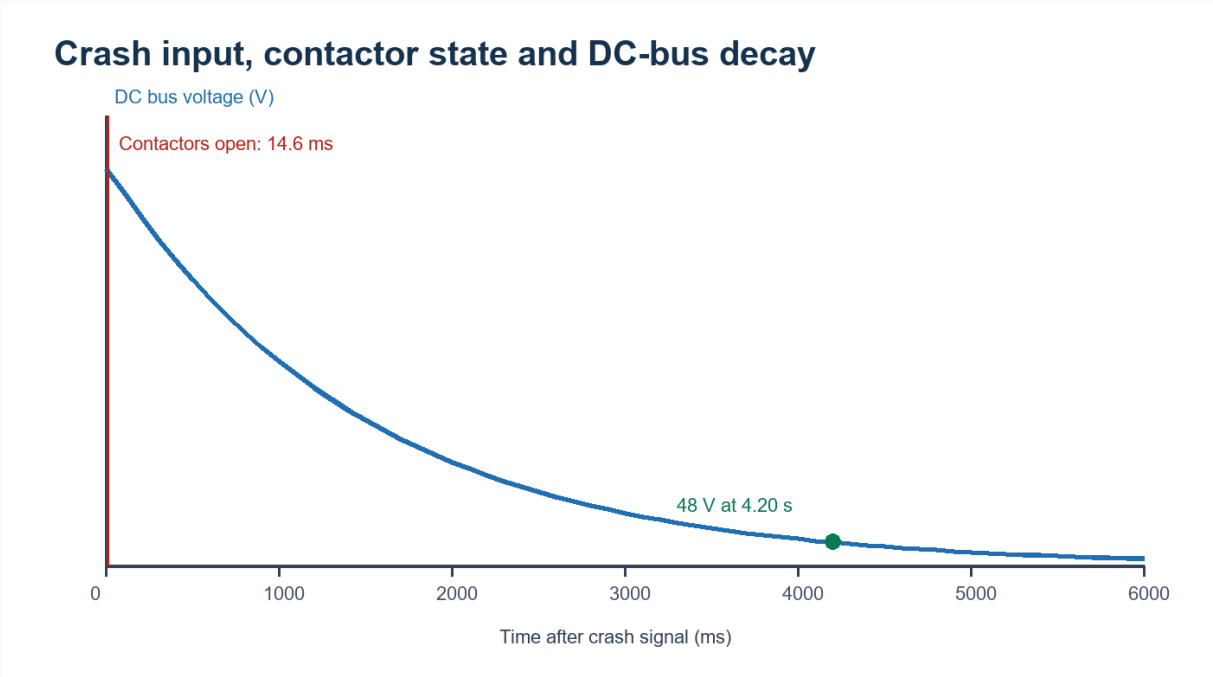


Figure 2. Recorded DC-bus trace; contactors opened at 14.6 ms and bus voltage reached 48 V at 4.20 s.

TR-ISO-03 MEASUREMENT RECORD

TR-ISO-03 - Post-impact isolation measurement

The following values were recorded exactly 60 seconds after the frontal-impact pulse. Calculation and acceptance are continued on page 5.

Table 5. Isolation measurement operands

Quantity	Symbol	Recorded value	Instrument
Elapsed time after impact	t	60.0 s	DAQ-14
DC-bus voltage	Vbus	402 V	DMM-22
Isolation resistance	Riso	1.26 Mohm	HI-07

Cross-page record. Do not decide the ohm-per-volt criterion from this page alone. The controlled formula, converted units and computed ratio appear on page 5.

TR-ISO-03 CALCULATION AND ALTERNATIVE PATH

TR-ISO-03 - Calculation and logic-path result

Controlled calculation. $R_{iso}/V_{bus} = 1,260,000 \text{ ohm} / 402 \text{ V} = 3,134 \text{ ohm/V}$. Acceptance threshold: $\geq 500 \text{ ohm/V}$. Result: PASS.

Table 6. MM-REQ-012 alternative paths

Alternative path	Evidence	Status
A - post-impact isolation ratio $\geq 500 \text{ ohm/V}$	3,134 ohm/V from pages 4-5	PROVEN
B - contactor opening $\leq 100 \text{ ms}$ after injected isolation fault	Isolation-fault injection not executed	UNTESTED

ANY_OF decision. Path A is proven; therefore the alternative post-impact protection requirement is satisfied even though Path B remains untested.

TR-TH-04 THERMAL AND COOLANT

TR-TH-04 - Thermal protection and coolant pressure

Table 7. Thermal/coolant results

Test	Measured result	Recorded disposition
Thermal shutdown heating ramp	Shutdown asserted at 87 deg C	PASS for shutdown
Recovery below 70 deg C for 60 s	Cool-down phase not run	NOT TESTED
Coolant pressure hold	2.00 bar for 30.0 min	PASS
Visible leakage during hold	0.00 mL observed through chamber window	PASS in-run observation

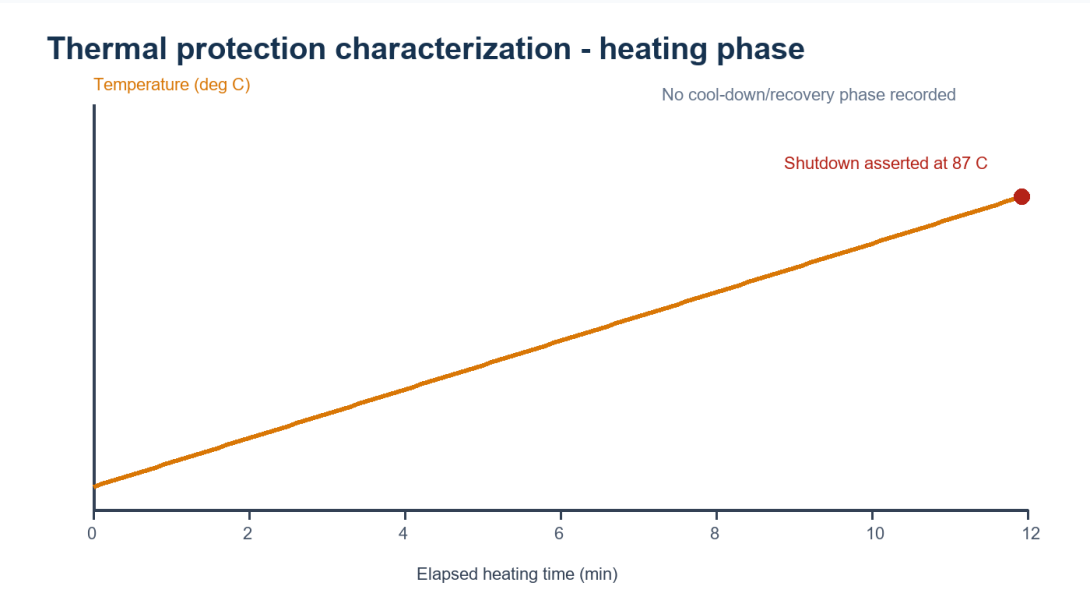


Figure 3. Heating-phase trace. It proves shutdown behavior but contains no recovery evidence.

TR-NW-05 COMMUNICATIONS AND OPEN WORK

TR-NW-05 - CAN endurance and deferred drive-away test

Table 8. CAN endurance summary

Duration	Messages expected	Messages received	Maximum interval
30.0 min	18,000	18,000	95.1 ms

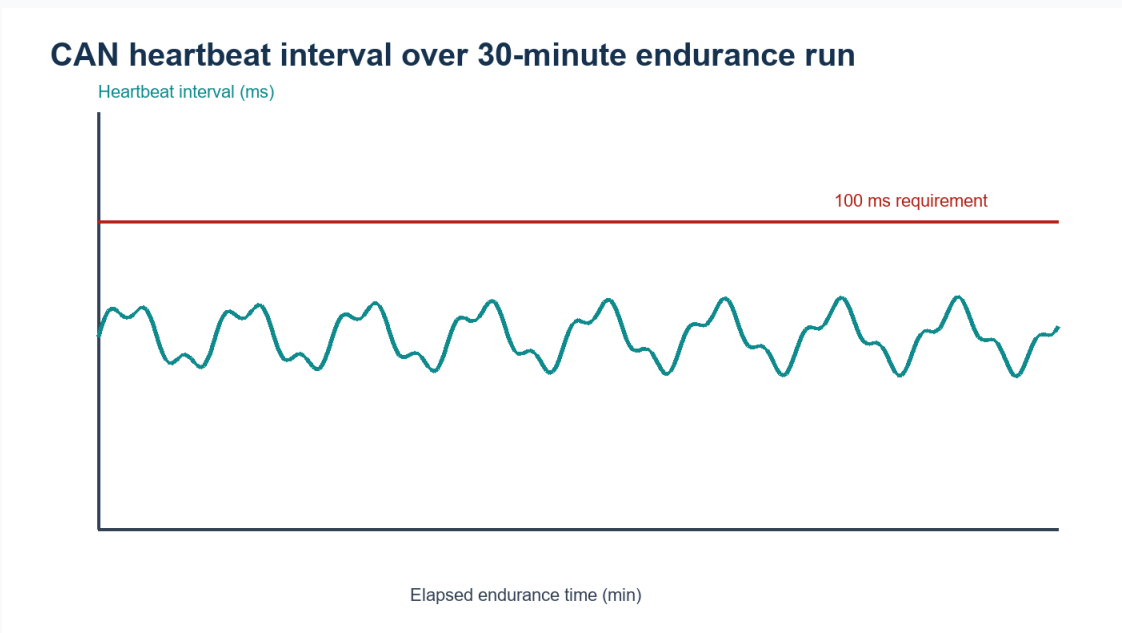


Figure 4. Heartbeat interval remained below 100 ms for the full 30-minute run; message counter showed zero loss.

Deferred test DT-17. Drive-away inhibition with the charge connector engaged was not executed. No torque or movement measurement is available.

TR-MECH-06 INGRESS AND INSTRUMENTS

TR-MECH-06 - J17 immersion and instrument register

Table 9. J17 immersion record

Parameter	Recorded value	In-run result
Water depth	1.00 m	Meets method
Immersion duration	30.0 min	Meets method
Electrical continuity after immersion	Normal	PASS
External visual check before disassembly	No moisture visible	PASS

Table 10. Acceptance-data instruments

Serial	Use	Calibration date shown in report
DMM-22	Voltage measurements	2026-02-12
DAQ-14	Timing and CAN data	2026-01-29
HI-07	Isolation resistance	2025-05-03

Inspection handoff. Rear-boot internal moisture and calibration-certificate validity were evaluated after test in AQL-QIR-221. This page alone does not close those questions.